

Breakthrough in quantum encryption

This is no exaggeration, but there's a ticking time bomb at the heart of modern computing platforms that enable our digital world. Quantum computers, often being heralded as the answer to so many of humanity's unfulfilled potentials, they're also harbingers of unprecedented doom. Which is why the finalisation of PQC (post-quantum cryptography) algorithms deserves greater attention, and how it will help us bolster global encryption standards.

Earlier in August 2024, NIST (National Institute of Standards and Technology) announced the final set of "encryption tools designed to withstand the attack of a quantum computer." This is a crucial step. Tasked with creating best standards and practices for the US government and its agencies, when NIST releases a set of new cryptographic standards designed to withstand cyberattacks from a quantum computer, its importance cannot be overstated enough.

Most of us have no tangible idea how different and mind-bogglingly quick quantum computers can be at a fundamental level compared to all the traditional computers we know and have heard of till date. They are insanely quick at certain tasks like factoring large integers and solving discrete logarithms, which are the mathematical foundations of many modern encryption protocols. For example, using Shor's algorithm, a quantum computer can factorise a large number exponentially faster than classical computers, effectively breaking RSA encryption. This capability poses a significant risk, as it could allow attackers to decrypt secure communications, financial transactions, and sensitive data with unprecedented speed. Imagine nothing is encrypted and private anymore, everything laid bare for cyberattackers to exploit. That's what's at stake, the very fabric of cryptographic systems that underpin our modern digital infrastructure, and their legitimacy in the quantum computing era.

But thankfully that worst case scenario is avoided due to NIST's approval of three PQC standards: ML-KEM (derived from CRYSTALS-Kyber), ML-DSA (derived from CRYSTALS-Dil-



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ithium), and SLH-DSA (derived from SPHINCS+) – three brand new post-quantum cryptography algorithms. ML-KEM is based on a key encapsulation mechanism selected for general encryption. ML-DSA is a lattice-based algorithm picked for its general-purpose digital signature protocols. SLH-DSA is a stateless hash-based digital signature scheme. According to IBM, both ML-KEM and ML-DSA were developed by IBM Research cryptography researchers in Zurich with external collaborators, and SLH-DSA was co-developed by a scientist who has since joined IBM Research.

There is, in fact, a fourth draft standard based on FALCON which is planned for finalisation by late 2024. The upcoming FALCON encryption is derived from NTRU-Lattice and will be a digital signature algorithm, according to the NIST.

It isn't just IBM leading the way, several other tech companies are moving quickly to safeguard their platforms and users from "harvest now, decrypt later" cyberattacks as well. Apple's iMessage platform's PQ3 has already begun implementing the ML-KEM encryption standard, so has Google Chrome with support for a hybrid KEM implementation. Known for its water-tight encrypted messaging, Signal chats are already secured by PQC. Zoom Workplace calls are also protected by PQC algorithms. The enabler of one-fourth of all connection requests across the internet, even Cloudflare has started turning on a version of ML-KEM encryption.

Apart from contributing to their co-development, the likes of Amazon, Google and Microsoft are also in the process of adopting at scale the latest PQC standards finalised by NIST. This shows how big tech enterprises are recognising the urgent need to safeguard and protect themselves with quantum-safe technology solutions.

Make no mistakes, the adoption of these new standards marks an essential step in safeguarding our data, ensuring that as quantum computing scales, it doesn't crash and burn everything about the digital world as we know it – that our encryption methods evolve to meet the challenge. **d**